Not a home assignment :)

This is a demo version of the midterm exam. Midterm will consist of 6 problems: two problems on binomial identities, two problems on recurrent equations, two problems on generating functions. Some problems will be similar to demo 

You may use one A4 cheat sheet in any form handwritten, printed, aquarelle painting...

1. You have 5 cups of different colors and 20 identical straws.
   (a) In how many ways you can place straws into cups?
   (b) In how many ways you can place straws into cups if at least 3 cups should be non-empty?

2. Simplify
$$\binom{5}{0}\binom{6}{4}+\binom{5}{1}\binom{6}{3}+\binom{5}{2}\binom{6}{2}+\binom{5}{3}\binom{6}{1}+\binom{5}{4}\binom{6}{0}.$$

3. Solve the recurrence equation
$$a_n=\frac{n(n+1)}{(n+2)(n+3)}a_{n-1},\quad a_0=100.$$
   Hint: find a_1, a_2, a_3 and guess the pattern.

4. You play a the following lottery with the casino. There is a hat on the table. A fair dice is thrown until the game ends.
   - If the dice shows one then you get $+1$ dollar immediately and the game ends.
   - If the dice shows two then the sum in the hat is multiplied by 2 and the game continues.
   - If the dice shows three then $+3$ dollars are added to the hat and the game continues.
   - If the dice shows four then the sum in the hat goes back to casino and the game ends.
   - If the dice shows five then the sum in the hat goes back to the casino and the game continues.
   - If the dice shows six then the game ends and you receive the content of the hat.

   Let e_n be your total expected payoff if there are n dollars in the hat now.

   Find the recurrence equation for e_n. You don't need to solve it.

5. Consider the recurrence equation
$$a_n-4a_{n-1}+5a_{n-2}=0 \text{ for } n\geq 2,\quad a_0=5, a_1=2.$$
   Find the generating function $g(t)$ for the sequence $(a_0,a_1,a_2,\dots)$.

6. The sequence $(b_0,b_1,b_2,\dots)$ has a generating function
$$g(t)=\frac{2}{1-3t}+\frac{5}{1-2t}.$$
   Find b_0, b_1 and b_{100}.